COMPUTER VISION TAKE-HOME · KRP SOLUTIONS

Sleep Detection from Video

Given an ordinary webcam video, my system says for every person and every second whether they are **awake**, **drowsy**, **asleep**, **unknown** or **absent**, lists their sleep intervals, and names each person from one reference photo. It uses only pretrained models plus a small set of explainable rules; nothing was trained.

Abhinand · B.Tech, 2nd year

Code: github.com/Abhinand-Nr/sleep-detection · 3-min demo video

0.95
asleep F1 on the held-out eval clips (precision 0.95, recall 0.95)

0 s
false "asleep" on the phone clip, vs 117 s for the naive eyes-closed rule

2 / 2
people named correctly in the two-person clip, from one photo each

9 clips
about 25 min of my own footage, labelled per second by hand

1 What counts as "asleep"

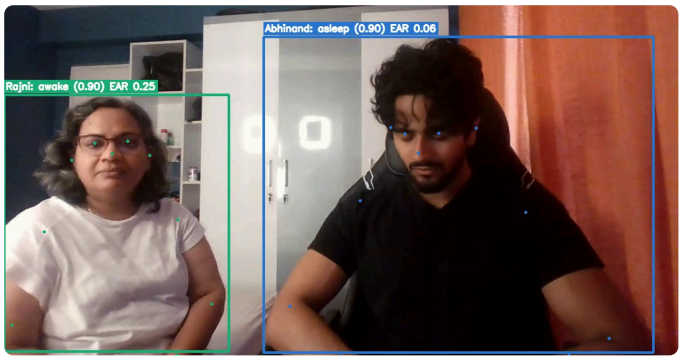
A camera cannot see brain activity, so I detect **behavioural sleep**: what a person watching the video would call asleep. The rule depends on what is visible, and every threshold below is a setting in `config.yaml`.

Face visible	Eyes closed (eye-openness below 0.5 × the person's own open-eye level) **and** body still for **10 s**	asleep · 0.9
Face hidden head on arms, facing away	Body still for **45 s** in a sleep posture: slumped, head on arms, or lying	asleep · 0.65
In between	Eyes shut longer than a 0.5 s blink but under 10 s, PERCLOS ≥ 0.3 over 30 s, or head nodding	drowsy
Not visible	Person mostly out of view or hidden: no guess is made	unknown

Why these choices. Ten seconds is far longer than a blink or a short eye rest, and requiring stillness as well stops "eyes closed while thinking" from counting. Without the eyes there is less evidence, so the body-only path waits longer and reports lower confidence. Waking up needs 2 s of open eyes or clear movement, so one bad frame cannot flip the state.

2 Signals, and why I chose them

People	YOLOv8n-pose (6.8 MB) finds people and 17 body keypoints; ByteTrack keeps one ID per person. Gives posture and a box to measure motion in.
Eyes	YuNet (0.2 MB) finds faces; MediaPipe Face Landmarker (3.8 MB) gives 478 face points and head angles. Eye aspect ratio (EAR) from 6 points per eye, calibrated on each person's first 10 s facing the camera. Cheap, explainable, and per-person calibration handles different eye shapes.
Head pose	Eyes are trusted only when the head is roughly frontal: turned < 35°, tilted down < 25° from normal and < 20° from the camera. Looking down at a phone makes open eyes look closed.
Motion	Pixel-change "motion energy" inside the person's box minus background change. Sleep means stillness; subtracting the background ignores light flicker and camera noise.
Identity	SFace 128-number face embeddings compared with `refs/`, voted over the whole track (threshold 0.363), with a clothing-colour backup. The name stays when the face is hidden.



Clip 8 at 0:45, system output. Abhinand is asleep (EAR 0.06), Rajni is awake (EAR 0.25). Names come from face matching: Abhinand's track scores 0.74 to his photo vs 0.07 to Rajni's; Rajni's two tracks score 0.78 / 0.76 vs 0.11 / 0.12.

Why pretrained models and rules, not a trained model

I had about 25 minutes of my own footage, which is enough to test rules but far too little to train a sleep classifier that I could trust. Rules built on well-tested pretrained detectors are easy to explain (each output row says *why*, e.g. "eyes closed + still 10 s") and easy to fix when they fail.

The pipeline runs in two passes: the slow model pass is cached (`signals.pkl`), and the fast rule pass can be re-run and tuned in seconds. There are 47 automated tests and a tested Dockerfile.

3 How I decided whether it works

Data. 9 clips (about 25 min, 1280×720, laptop webcam, one room). Eight have one person: normal work, 5 s eye closures, acted sleep facing the camera (×2), head on desk, sideways on a phone, looking down at a phone, and dim light. One clip has two people (Abhinand and Rajni, 3.5 min). Every second was labelled by hand; closures under 10 s are labelled drowsy.

Split. Tune = clips 1, 3a, 4a (486 s). Only two thresholds were grid-searched there (24 combinations): `closed_ratio` 0.6 → 0.5 and `still_threshold` 0.01 → 0.02. **Eval** = clips 2, 3b, 4b, 5, 6, 8 (1161 person-seconds), scored once with those values.

Baselines. Naive: eyes below threshold = asleep, no timing. **Face-only**: my state machine without the body signals. **Full**: everything above. **Metrics** are per second with asleep as the positive class: precision, recall, F1, IoU of asleep seconds, plus accuracy over all five states.

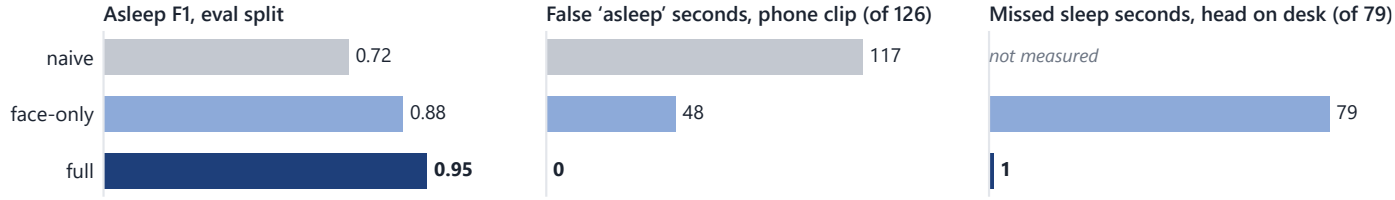
Results on the eval split (asleep = positive class)

system	precision	recall	F1	IoU	accuracy, all states
Naive	0.58	0.96	0.72	0.57	0.80
Face-only	0.82	0.96	0.88	0.79	0.77
Full system	**0.95**	**0.95**	**0.95**	**0.91**	**0.78**

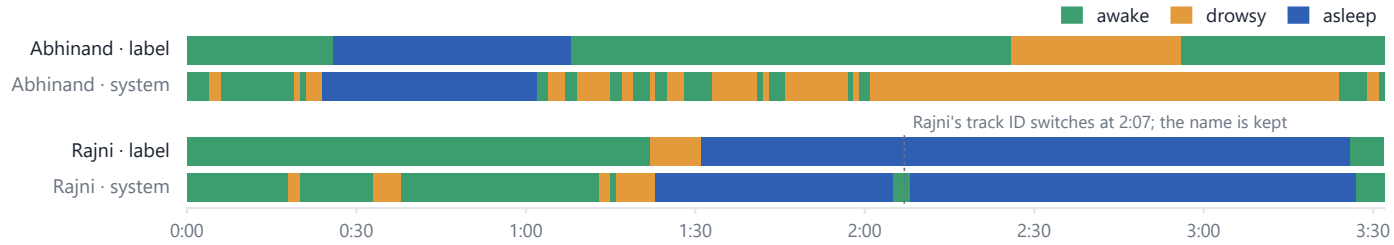
1161 person-seconds never used for tuning. Tune split, full system: F1 0.96. Accuracy over all five states is held down by extra "drowsy" (§4).

What each part adds, clip by clip

On eyes-closed sleep facing the camera the full system reaches F1 0.98 (clip 3b), and 0.95 in dim light at 16 fps (clip 6). All three 5 s eye closures in clip 2 were called drowsy, not asleep (naive: 15 false asleep seconds). With two people (clip 8) F1 is 0.94 and both are named correctly. Speed is 5–9 processed frames/s on a laptop CPU with no GPU, i.e. 1–2.2× the video length.



Each step removes a class of error. Adding timing rules (face-only) and then body signals (full) takes the phone clip from 117 to 48 to 0 false asleep seconds. With the face hidden on the desk, face-only misses all 79 sleep seconds; the full system misses 1, using the body path.



Clip 8, labels vs system, per second. Both sleep periods are found, and Rajni's 115 s sleep is followed across a tracker ID switch. The extra orange on Abhinand's row is the over-eager nodding rule described in §4.

How the final numbers were reached (please read before trusting them)

The first evaluation gave eval F1 **0.61** (kept in `outputs/evaluation_v1_before_fixes/`). Looking at the eval failures, I found three bugs and fixed each with a new test:

1. The per-event IoU metric mixed clips together, giving values above 1.
2. The phone clip started head-down, so "head down" was learned as normal (74 false asleep seconds). Now calibration uses only camera-facing frames, plus a 20° absolute head-down limit.
3. Rajni's track ID changed at 2:07 with 0.4 s of overlap, so the new track was left unnamed. Up to 1 s of overlap is now allowed at a handover.

Four label spans in clips 4b and 8 disagreed with the video; the actor re-checked and corrected them (clip 4b was really phone use, not sleep). Both steps used eval clips, so the final 0.95 is **optimistic**. The tune-split F1 (0.96) did not change.

4 Where it fails, and what I would do next

FALSE POSITIVE RISK **Phone use vs dozing**

Head pitch was +15° to +31° while looking at a phone and about +18° when dozing with the chin dropped. The 20° limit separates them on my clips, but the margin is only a few degrees.

UNTESTED **Sleep while facing away**

No eval clip has it (clip 4b turned out to be phone use), so the body-only path is only tested on clip 4a, a tune clip. In clip 8, sleep onset was found about 8 s before the label (Rajni's eyes closed at 1:22).

MAIN ERROR **Too much "drowsy"**

225 of 794 labelled-awake eval seconds were called drowsy, mostly the head-nodding rule firing on normal head movement. It does not touch the asleep metrics, but keeps all-state accuracy at 0.78.

SCOPE **Small, acted data; offline**

Acted sleep, two people, one room, one camera. The offline "backfill" marks the first 10 s of a sleep once it is confirmed, so a live version would report onset 10 s late.

Next, with more time: record more people, rooms and real (not acted) naps, including more eval clips facing away; replace the nodding rule with a frequency-based one or require repeated dips; use YOLO's "cell phone" class and hand position to separate phone use from dozing; add a live webcam mode (same rules, no backfill).

5 Long-term vision

One explainable state per person per second can grow into a monitoring tool for places where falling asleep matters, with video kept on the device.

NOW
Offline, explainable
- Recorded video, up to two named people
- Per-second CSV, intervals, annotated video
- Every threshold in one config file

NEXT
Live and context-aware
- Live webcam mode with running identity votes
- Phone and hand detection to tell phone use from dozing
- Better nodding rule; more, real sleep data

LATER
Real settings
- Classrooms, offices, control rooms, driver monitoring
- Several cameras with re-ID across them
- Alerts (e.g. asleep for N minutes) and a dashboard of timelines

LONG TERM
Private and learned
- Edge devices: only states and timestamps leave the camera
- A small learned model once real labelled data exists, with rules kept as the explanation
- Opt-in faces only; clear consent